

OpenClaw Temperature and Humidity Reading Tutorial

Before starting, please complete the following in the `Pre-use Preparation` directory: [01-Peripheral Hardware Configuration], [02-OpenClaw Ajetonson-KEY Configuration], [03-OpenClaw Model Switching], [04-AI Voice Interaction Configuration], [05-Three Dialogue Scheme Configuration Test]. If you are currently only using TUI / WhatsApp, you can skip `04` for now. This article assumes the basic environment is already prepared and only covers DHT-related content.

1. Tutorial Objective

This article demonstrates OpenClaw's natural language control capabilities through the DHT temperature and humidity sensor. After completing this tutorial, readers will be able to:

- Understand the basic approach of OpenClaw reading temperature and humidity
- Interact with OpenClaw through three methods: WhatsApp, TUI, and voice interaction
- Complete four DHT cases: single reading, comfort judgment, threshold alert, and OLED linked display
- Understand the complete chain from "user question" to "sensor reading" and "result feedback"

2. Hardware Preparation

- OpenClaw main control board
- DHT temperature and humidity sensor `[Default DHT11]`
- `[Optional]` OLED display
- `[Optional]` Microphone module
- Wiring diagram



3. Software Preparation

Code entry points used in this tutorial

- Low-level driver code: `/home/jetson/hardware_hal_demo/`
- Temperature and humidity reading command: `/home/jetson/hardware_hal_demo/dht`
- Temperature and humidity linked display: `/home/jetson/hardware_hal_demo/dht_oled`

- Voice entry: `/home/jetson/voice_to_openclaw/src/main.py`

The `dht` tool supports the following capabilities:

- Read current temperature and humidity
- Output results in `json` or `text` format
- Use `mock` fake data for local debugging

```
dht_oled #Display temperature and humidity data on OLED
dht #Return `json` result
```

If you want to display temperature and humidity results directly on the OLED, you can also use `dht_oled`.

Note: If you want to write your own script to read temperature and humidity data using the `ajetson` Python driver library, it is recommended to read DHT data at 2-second intervals, as the DHT sensor works best with a 2-second sampling period.

4. Testing Temperature and Humidity Reading

Before starting this tutorial, please complete a basic `openclaw tui` dialogue self-check first; if you haven't done so, see [05-Three Dialogue Scheme Configuration Test]. After confirming that OpenClaw can reply normally, proceed to the DHT-specific test.

Enter the following command in the terminal to enter TUI:

```
openclaw tui
```

```

[OpenClaw 2026.3.12 (6472949)
  Your terminal just grew claws--type something and let the bot pinch the busywork.

openclaw tui - ws://127.0.0.1:18789 - agent main - session main
connecting | idle
agent main | session main | unknown | tokens ?

18:28:21 [plugins] feishu_chat: Registered feishu_chat, feishu_chat_members
18:28:21 [plugins] feishu_im: Registered feishu_im_user_message, feishu_im_user_fetch_resource, feishu_im_u
ser_get_messages, feishu_im_user_get_thread_messages, feishu_im_user_search_messages
18:28:21 [plugins] feishu_search: Registered feishu_search_doc_wiki
18:28:21 [plugins] feishu_drive: Registered feishu_drive_file, feishu_doc_comments, feishu_doc_media
18:28:21 [plugins] feishu_wiki: Registered feishu_wiki_space, feishu_wiki_space_node
18:28:21 [plugins] feishu_sheets: Registered feishu_sheet
18:28:21 [plugins] feishu_im: Registered feishu_im_bot_image
18:28:21 [plugins] Registered all OAPI tools (calendar, task, bitable, search, drive, wiki, sheets, im)
18:28:21 [plugins] feishu_doc: Registered feishu_fetch_doc, feishu_create_doc, feishu_update_doc
18:28:21 [plugins] feishu_oauth: Registered feishu_oauth tool

session agent:main:main
gateway connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens 0/1.0m (0%)

```

You can try entering the following content in the terminal:

- Read the current temperature and humidity
- What is the current ambient humidity
- Help me judge whether the current environment is comfortable
- Alert me if the temperature is too high

```
Read current temperature and humidity

Current temperature and humidity readings:
- Temperature: 29.9°C (85.8°F)
- Humidity: 52%

Sensor: DHT11 ✓

What is the current environmental humidity

Current environmental humidity: 52%

(Temperature is 29.9°C if you need that too) 📺
connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens
48k/1.0m (5%)
```

5. Three Schemes for Communicating with OpenClaw

Scheme	Suitable Scenarios	Advantages	Recommended Use
WhatsApp	Remote control, classroom demonstration	Easy to connect, suitable for multi-person experience	Demonstrate "remote natural language control"
TUI	Local debugging, command verification	Most direct, easiest to troubleshoot	First run through the control chain
Voice code	Voice interaction, complete AI experience	Closest to a real conversational experience	As the final demonstration scheme

If you have the optional AI voice module + speaker, Feishu dialogue, TUI, and WhatsApp dialogue need to play back OpenClaw's replies via TTS. You can run the following program in the terminal. The AI voice module dialogue does not need to run this repeatedly.

- Modify the TTS toggle parameter

```
/home/jetson/voice_to_openclaw/.env
```

```
ENABLE_TTS=true #This parameter controls whether to broadcast. After modifying and saving, run the following program in the terminal
```

- Enter in terminal

```
cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts
```

```
(.venv) jetson@yahboom:~/voice to openclaw$ python -m src.openclaw_session_tts  
[系统] OpenClaw 会话播报器已启动  
[系统] 监听目录=/home/jetson/.openclaw/agents/main/sessions  
[系统] 监听渠道=feishu,tui,whatsapp
```

5.1 WhatsApp Scheme

For WhatsApp connection steps, please directly refer to [05-Three Dialogue Scheme Configuration Test]. This section only includes dialogue examples suitable for the DHT tutorial.

After creation, you can directly converse, for example:

- Read the current temperature and humidity
- Judge whether it's a bit hot right now
- Alert me if the humidity is too low

19:53



< 2



+86 153 3885 7526 (你)

给自己发消息

OLED

19:52 ✓✓

Done! "welcome to openclaw" is now displayed on the OLED screen. The text was split across two lines due to the 128-character width limit of the display.

19:52 ✓✓

Read current temperature and humidity

19:53 ✓✓

Current readings from the DHT11 sensor:

- **Temperature:** 29.7°C (85.5°F)
- **Humidity:** 52%

The environment is fairly warm and moderately humid.

19:53 ✓✓



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space

return



5.2 TUI Scheme

If you have completed the TUI self-check in Part 0, simply enter the following command in the terminal to continue the conversation:

```
openclaw tui
```

```
❯ OpenClaw 2026.3.12 (6472949)
  Your terminal just grew claws-type something and let the bot pinch the busywork.

openclaw tui - ws://127.0.0.1:18789 - agent main - session main
connecting | idle
agent main | session main | unknown | tokens ?

18:28:21 [plugins] feishu_chat: Registered feishu_chat, feishu_chat_members
18:28:21 [plugins] feishu_im: Registered feishu_im_user_message, feishu_im_user_fetch_resource, feishu_im_
ser_get_messages, feishu_im_user_get_thread_messages, feishu_im_user_search_messages
18:28:21 [plugins] feishu_search: Registered feishu_search_doc_wiki
18:28:21 [plugins] feishu_drive: Registered feishu_drive_file, feishu_doc_comments, feishu_doc_media
18:28:21 [plugins] feishu_wiki: Registered feishu_wiki_space, feishu_wiki_space_node
18:28:21 [plugins] feishu_sheets: Registered feishu_sheet
18:28:21 [plugins] feishu_im: Registered feishu_im_bot_image
18:28:21 [plugins] Registered all OAPI tools (calendar, task, bitable, search, drive, wiki, sheets, im)
18:28:21 [plugins] feishu_doc: Registered feishu_fetch_doc, feishu_create_doc, feishu_update_doc
18:28:21 [plugins] feishu_oauth: Registered feishu_oauth tool

session agent:main:main
gateway connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens 0/1.0m (0%)
```

5.3 Voice Code Scheme

Requires optional AI voice interaction module + speaker

For voice module installation, environment configuration, and basic wake-up process, please directly refer to [04-AI Voice Interaction Configuration].

You need to say "Hi, yahboom" to wake up for conversation.

After wake-up, you can directly speak control commands, for example:

- What is the current temperature
- Help me read the humidity
- Tell me if it's too stuffy

```
[系统] ASK=xunfei_ws/1at/en_us | 发布-gateway_chat | TTS=开/xunfei_ws | 唤醒-串口
[系统] 串口唤醒已启用: /dev/ttyUSB0@115200
[提示] 说“清空上下文”可以重置当前会话
[唤醒] 检测到串口信号, 开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 9990 ms 语音, 正在转写

[你] Read current temperature and humidity.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200

[OpenClaw] Current readings from the DHT11 sensor:

- **Temperature:** 30°C
  (86°F)
- **Humidity:** 51%

The sensor is working well! 🌡️💧
[信息] 回复摘要失败, 已退回本地清洗播报
[唤醒] 检测到串口信号, 开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 6810 ms 语音, 正在转写

[你] It's too hot. Please, let me know.
[发送] 正在交给 OpenClaw
```

6. Comprehensive Cases

Note: The following are all demonstrated using the `openclaw_tui` method. You can also choose WhatsApp or voice methods to complete them.

6.1 Case 1: Single Reading

Experimental Objective

Obtain a single reading of the current temperature and humidity to complete the most basic environmental sampling.

Effect Description

After OpenClaw receives the instruction, it will call the DHT driver to read the current temperature and humidity, and return the results to the user in natural language.

Example Instructions

- Read the current temperature and humidity
- What is the current indoor temperature
- Help me check the current humidity

6.2 Case 2: Comfort Judgment

Experimental Objective

Have OpenClaw make a simple judgment about the current environment after reading the data.

Effect Description

In addition to just reporting numbers, OpenClaw can also roughly judge states like "too hot", "too cold", "too dry", or "too stuffy" based on temperature and humidity, making the output sound more like a natural conversation.

Example Instructions

- Help me judge whether the current environment is comfortable
- Is the current temperature and humidity suitable for long-time office work
- Give me a brief evaluation based on the temperature and humidity

6.3 Case 3: Threshold Alert

Experimental Objective

Have OpenClaw alert the user when the temperature or humidity exceeds a set range.

Effect Description

When the temperature is too high or too low, or the humidity is too high or too low, OpenClaw can directly output alert messages, or link with RGB lights for more obvious feedback.

Example Instructions

- Alert me if the temperature exceeds 30 degrees
- Notify me if the humidity drops below 40%
- Has the alarm condition been reached

Extensible Ideas

- Link with RGB for red-yellow-blue three-color alerts
- Link with OLED to display alarm text
- Add a buzzer for stronger alerts

6.4 Case 4: OLED Linked Display

Experimental Objective

Display the just-read temperature and humidity results directly on the OLED, forming a complete data display chain.

Effect Description

After OpenClaw completes an environmental sampling, it can not only reply the results to the user but also simultaneously write the temperature and humidity to the OLED screen, forming a complete "reading + display" effect.

Example Instructions

- Read the current temperature and humidity and display on OLED
- Write the temperature and humidity to the screen
- Create a simple temperature and humidity display page

Extensible Ideas

- Periodically refresh OLED content
- Create a temperature and humidity dashboard
- Link with a weather station project to form a complete application